



PX4 flight management unit (FMU) I/O kit

electronic design automation tool) for schematic capture and PCB layout.

TeleMetrum is recording dual-deploy altimeter for high-power model rocketry with integrated GPS and bi-directional telemetry link. It features a 70cm ham-band receiver for telemetry downlink and integrated support for electronic ignition of pyro stages.

TeleMega is a version of TeleMetrum, which is designed for high-power rockets. Although it sounds like it uses an ATmega chip, it actually runs on ST Micro's MCU. In addition to the features of TeleMetrum, TeleMega also makes it possible to configure pyro events with up to six pyro channels, and can be based on time and various flight events as well as status-like angles from vertical and more.

TeleMega firmware runs on an operating system called AltOS, which provides features like flight monitoring, post-flight analysis, device configuration and firmware update. AltOS, through which these features can be accessed, runs on Linux, Mac OS X and Windows as well.

Rocket Mega

Designed by OSAerospace to be a low-cost open source platform for development and experimentation in rocketry and other aerospace applications, the Rocket Mega project is based on Arduino hardware.

Major contributors to this report

- Anmol Agrawal, UX/UI designer, Interaction-Design
- Darshan Virupaksha, systems engineer, Altux Innovations and co-organiser of IoTBLR
- Eben Upton, founder, Raspberry Pi Foundation
- Ganesh Kumar, country sales manager, Atmel
- Lynnette Reese, technical team lead, Mouser Electronics
- Selvaraj Kaliyappan, vice president - PES Engineering, Mistral Solutions



Rocket Mega revision 2

What it essentially contains is a Rocket Mega shield, code and an Eagle CAD library that contains custom parts and footprints. Complete details on how to build the board, including bill of materials and schematics are available at opensource.aerospace.com

The R1 version is a shield designed to work with Arduino Mega board. The R2 version comes with several improvements, yet still manages to keep the same price point in raw parts. Improvements include using MOSFETs rather than relays for controlling the pyro channels, apart from shifting to ceramic capacitors, improved routing and fewer individual parts.

TI's MSP430 LaunchPad + CC430 kit rocket launch controller

TI's aptly named LaunchPad development kit combined with a CC430 low-power wireless development kit provides for a good base to build a remove model rocket launch controller with streaming video support.

Digi-Key website's video library has a good video showing the system in action, where CC430 controls the current to the rocket ignition system by using an optically-isolated

MOSFET. Another laptop-connected kit controls a servo that supports video equipment.

Autopilot

The APM Autopilot suite is an open source autopilot solution that includes hardware, firmware and software to run an autopilot vehicle. The firmware

is what contains the skill-set code, which configures the system for the kind of vehicle that you are going to be running. Examples are ArduPlane that allows you to give a fixed-wing plane full autonomous capability and ArduRover that does the same for an unmanned ground vehicle. APM's APM:Rover won the 2013 and 2014 Sparkfun Autonomous Vehicle Competition too, so that gives good credibility to the system.

The hardware that you can opt from for this system includes 3DR Pixhawk, which features an advanced processor and sensor technology from STMicro-electronics and comes with a NuttX real-time operating system (RTOS).

APM 2.6 is another piece of hardware that can be integrated for implementing autopilot features onto the board. It is Arduino compatible but requires a GPS unit and a compass module for full autonomy.

One small step

Do not wait. You are in the golden age for developers, with access to a ton of code and hardware that has already enabled others to develop amazing projects. Start tinkering with these boards to build your dream project. Hopefully, one of these might let you touch base with space: the final frontier. ●